Overview of .NET

1. What is .NET?

.NET is a **free, open-source, cross-platform development platform** created by Microsoft for building different types of applications.

Developers can create:

- Web applications
- Desktop applications
- Mobile apps
- Cloud services
- APIs and microservices
- IoT applications
- Games

It supports multiple programming languages such as:

- **C#**
- **F#**
- **VB.NET**

Among these, **C#** is the most commonly used language.

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2. Evolution of .NET

1️⃣ .NET Framework (2002)

The original version of .NET designed for **Windows-only applications**.

Used for:

- Windows Forms
- WPF
- ASP.NET
- Web Services

Limitations:

- Works only on Windows
- Not optimized for cloud or microservices

2️⃣ .NET Core (2016)

A redesigned version of .NET created to solve the limitations of .NET Framework.

Features:

- Cross-platform (Windows, Linux, macOS)
- High performance
- Lightweight
- Open source

Common frameworks built on it:

- ASP.NET Core
- Entity Framework Core

3️⃣ Modern .NET (.NET 5+)

Starting from **.NET 5**, Microsoft unified the platform.

Versions:

- .NET 5
- .NET 6 (LTS)
- .NET 7
- .NET 8 (Current LTS)

This unified platform supports **web, cloud, mobile, desktop, and AI development**.

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3. .NET Architecture

The .NET platform consists of three major components.

1️⃣ Common Language Runtime (CLR)

The **CLR** is the **execution engine of .NET**.

Responsibilities:

- Memory management (Garbage Collection)
- Exception handling
- Security
- Thread management
- Code execution

Flow:

C# Code

↓

Intermediate Language (IL)

↓

CLR

↓

Machine Code (JIT Compiler)

2️⃣ Framework Class Library (FCL)

The **FCL** is a large collection of **pre-built classes and APIs**.

Examples:

Namespace	**Purpose**
System	Basic classes
System.IO	File operations
System.Collections	Collections

Namespace	Purpose
System.Data	Database access
System.Net	Network programming

Example:

```
using System;


class Program
{
    static void Main()
    {
        Console.WriteLine("Hello .NET");
    }
}
```

Here **Console class** comes from **FCL**.

3️⃣ Languages

.NET supports multiple programming languages.

Examples:

Language	Purpose
C#	Most widely used
F#	Functional programming
VB.NET	Legacy enterprise apps

All these languages compile into **Intermediate Language (IL)** which runs on CLR.

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4. .NET Application Types

Developers can build multiple types of applications.

Web Applications

Framework:

- ASP.NET Core

Examples:

- REST APIs
- MVC web applications
- Microservices

Desktop Applications

Frameworks:

- Windows Forms
- WPF

Mobile Applications

Framework:

- .NET MAUI

Allows building Android, iOS, Windows apps using **single codebase**.

Cloud Applications

.NET integrates strongly with cloud platforms like:

- Microsoft Azure

Used for:

- Microservices
- Containers

- Serverless applications

Gaming

Game development using:

- Unity (uses C#)

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5. .NET Development Tools

.NET SDK

Software Development Kit used to build and run applications.

Commands example:

dotnet new webapi

dotnet build

dotnet run

IDEs

Popular development environments:

- Microsoft Visual Studio
- Visual Studio Code
- JetBrains Rider

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6. Advantages of .NET

1️⃣ **Cross Platform**

Run on:

- Windows
- Linux
- macOS

2️⃣ High Performance

.NET uses:

- JIT compilation
- Optimized runtime

This makes it suitable for **high-performance APIs and microservices**.

3️⃣ Large Ecosystem

Includes frameworks such as:

- ASP.NET Core
- Entity Framework Core
- Blazor

4️⃣ Strong Security

Features:

- Code access security
- Authentication and authorization support
- Data protection APIs

5️⃣ Excellent Tooling

Visual Studio provides:

- Debugging
- Code analysis

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7. .NET Execution Process

The execution flow of a .NET application:

Step 1: Write Code (C#, F#, VB.NET)

Step 2: Compile

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Intermediate Language (IL)

Step 3: CLR Execution

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JIT Compiler converts IL to Machine Code

Step 4: Application Runs

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8. Where .NET is Used in Industry

Used in:

- Enterprise web applications
- Banking systems
- Cloud APIs
- Microservices
- SaaS platforms

Large companies using .NET include:

- Stack Overflow
- Dell
- Microsoft